

Xuhao Luo

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Education

University of Illinois Urbana-Champaign Ph.D. Candidate in Computer Science	Aug. 2021 - Nov. 2026 (Anticipated)
University of California San Diego M.S. in Computer Science	Sep. 2019 - Mar. 2021
University of Science and Technology of China B.S. in Applied Physics	Sep. 2015 - Jun. 2019

Interests

Storage and Memory Systems for AI (High-performance storage, Multi-tier KV-Cache, GPUDirect storage, etc.)

Research Publication

- **Xuhao Luo**, Emaan Atique, Aishwarya Ganesan, Ramnatthan Alagappan, **SkyLog: Reducing Cost and Latency in a Multi-Cloud Shared Log** (In submission to NSDI)
- **Xuhao Luo**, Shreesha G. Bhat*, Jiyu Hu*(*equal contribution), Aishwarya Ganesan, Ramnatthan Alagappan, **LazyLog: A New Shared Log Abstraction for Low-Latency Applications (SOSP 2024)** *Best Paper Award, Invited to ACM ToCS*
- **Xuhao Luo**, Ramnatthan Alagappan, Aishwarya Ganesan, **SplitFT: Fault Tolerance for Disaggregated Datacenters via Remote Memory Logging (EuroSys 2024)**
- **Xuhao Luo**, Weihai Shen, Shuai Mu, Tianyin Xu, **DepFast: Orchestrating Code of Quorum Systems (USENIX ATC 2022)**
- Shreesha G. Bhat, Tony Hong, **Xuhao Luo**, Jiyu Hu, Aishwarya Ganesan, Ramnatthan Alagappan, **Low End-to-End Latency atop a Speculative Shared Log with Fix-Ante Ordering (OSDI 2025)**
- Zhiyuan Guo*, Yizhou Shan*(*co-first author), **Xuhao Luo**, Yutong Huang, Yiyang Zhang, **Clio: A Hardware-Software Co-Designed Disaggregated Memory System (ASPLOS 2022)**

Experience

Google <i>Software Engineer Intern, Host: Xiaozhou (Steve) Li</i>	May. 2026 - Aug. 2026 <i>Sunnyvale, CA, USA</i>
· Designed and implemented a general framework to extend Bigtable compaction with user-defined functions. Deployed in production and demonstrated its feasibility to replace certain data processing pipeline.	
Amazon Web Service <i>Applied Scientist Intern, Mentor: Prof. George Amvrosiadis, Visiting Scholar</i>	May. 2024 - Aug. 2024 <i>Seattle, WA, USA</i>
· Investigated and improved ShardStore (storage system used by AWS S3) reclamation, and built a tool to evaluate different reclamation policy.	
Amazon Web Service <i>Applied Scientist Intern, Mentor: Shen Li, Principle Engineer</i>	May. 2022 - Aug. 2022 <i>Seattle, WA, USA</i>
· Improved the performance and reliability of the volume metadata updating workflow for AWS S3 volume metadata cache service with Amazon Quantum Ledger Database (QLDB).	
University of Illinois Urbana-Champaign <i>Research Assistant</i>	May. 2021 - Now <i>Urbana, IL, USA</i>
· <i>SkyLog</i> : Built a shared log abstraction for multicloud applications, with seamless reconfiguration and reduced egress cost.	
· <i>LazyLog</i> : Built a new shared log system that offers low-latency appends for latency-critical applications by lazily ordering log entries across multiple storage shards.	
· <i>SplitFT</i> : Built a new fault-tolerant approach for storage-centric cloud applications by replicating WALs on remote nodes using RDMA.	
· <i>DepFast</i> : Built a framework using C++ coroutine to implement and reason about fail-slow tolerant distributed systems in an easy and effective way.	
Microsoft Research <i>Research Intern</i>	Jun. 2020 - Sep. 2020 <i>Beijing, China</i>
· Designed and implemented a task scheduling and dispatching system for distributed ML, and a CUDA-based high-performance inter-GPU communication channel for distributed ML.	
· Multi-GPU collective operation(AllReduce, AllGather, Broadcast) throughput outperforms NCCL by at most 18.4%.	

University of California San Diego
Research Assistant, advised by Prof. Yiyang Zhang

Sep. 2019 - Dec. 2020
La Jolla, CA, USA

- Implemented a hardware/software co-designed disaggregated memory with FPGA. Worked on the network stack design and implementation.

University of Science and Technology of China
Graduation Project, Advisor: Prof. Xi Jin

Jan. 2019 - May. 2019

- Designed and implemented a general 8-bit quantized convolution module on Xilinx Virtex FPGA.
- Developed the TensorFlow C++API for the hardware accelerator using OpenCL. On ResNet-50 CNN, achieved a speedup of 5.17x and a memory usage reduction of 66% compared with the CPU TensorFlow implementation on Xeon E5 2686.

Skills

Language	C/C++, Python, Go, Java, Rust, Haskell, Verilog
Tools/Framework	RDMA, vLLM, SGLang, TensorFlow, Docker, Zookeeper, LLVM

Honors and Awards

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| • SOSP'24 Best Paper Award | Nov 2024 |
| • SOSP'24 Student Travel Grant | Nov 2024 |
| • EuroSys'24 Student Travel Grant | Apr 2024 |
| • ASPLOS'22 Student Travel Grant | Feb 2022 |
| • USTC Class of 2019 Outstanding Graduates | May 2019 |

Services

- EuroSys'26 Shadow PC
- OSDI'23 Artifact Evaluation Committee
- USENIX ATC'23 Artifact Evaluation Committee
- ACM TAAS (Transactions on Autonomous and Adaptive Systems) Invited Reviewer